

Audit and Replace Shelf Labels SOP

One episode replaces one worn label at one shelf position. It runs from a shelf strip lying in its cradle with a worn label in the position, a fresh label sheet with the corner of the next label lifted, an empty audit box and the wand on its spot, to a position carrying one fresh label lying flat and square, the wand beeped and green, the worn label lying in its own numbered box on the audit sheet, and every tool back on its spot. Twelve episodes clear one strip. One strip is one session.

All of this happens at the bench. The strip is taken off the shelf and brought to the bench before the session, and it goes back on the shelf after the session. Both of those moves happen off camera. No label is ever peeled or applied while the strip is on the shelf.

The order never changes: look at the position, peel the worn label off, lay the worn label in its box, peel one fresh label, place it square, press it flat, scan it. No fresh label is peeled until the position is bare. The fresh label is placed and pressed while the strip lies in its cradle, so the strip is never held in the air while a label goes on it.

The gripper does all the work. It peels the worn label, lays it in its box, peels the fresh label off the sheet, places it, presses it, and holds the wand over it to scan it. The cradle holds the strip still while the labels come off and go on, so the gripper is free for the label. The cradle is the only thing that ever holds the strip still, and neither the cradle nor the strip is moved during a session.

Only one thing is held at a time. Nothing is passed from one grip to another, no grip is changed while an object is off the table, and nothing is dragged or slid along the table. Every pick lifts straight up off the surface before it moves anywhere.

The positions are worked in order, left to right along the strip, one position per episode, position 1 first. The counts are fixed: one worn label off and one fresh label on per episode, one press of 2 seconds, up to three scan tries, and twelve positions to a strip. There is no second fresh label laid over a first one, and no extra press once the wand has been picked up. A label that lands on the table is left exactly where it falls, tagged, and the episode goes on. Nothing is ever tidied up on camera.

Setup

Finish both checklists before starting an episode.

Cell configuration

Environment camera: 90 mm.

cell_type: single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole bench top in one shot: the fresh label spot, the wand spot, the cradle with the whole strip in it, the audit sheet spot and the waste spot.
3. The position stays in frame from the moment the worn label is touched until the new label is pressed, and the new label and the wand window both stay in frame for the whole scan.
4. The gripper is at home and open.
5. The bench top is dry and holds nothing but the items in the Materials checklist.
6. The wand is on. It gives one beep and a green light on a test read of a spare fresh label.
7. The gripper reaches the fresh label spot, the wand spot, all twelve positions on the strip, all twelve boxes on the audit sheet and the waste spot without stretching to a joint limit or folding in on itself. Bring a spot forward until this is true.
8. The paths from the fresh label spot to the strip, from the strip to the audit sheet, and from the wand spot to the strip are clear. Carry the wand along its path by hand during setup: it must clear the cradle walls and pass over no other position than the one being worked. Widen the gaps until it does.

Materials checklist

1. One shelf strip lies in the cradle in the middle of the bench, face up, with all twelve positions in view and the position numbers readable.
2. Each of the twelve positions carries one worn label at the start of the session. A position with a sound label is a reset error, not something to replace.
3. Each position is a flat, dry, unbroken patch on the strip, so a fresh label lies down on it with nothing under the label. The patch has a printed border the label sits inside.
4. The cradle grips the bench. Push it by hand during setup: if it slides or lifts, weigh it down or clamp it. Nothing pins the cradle during the episode.
5. The cradle is open at the top and holds the strip flat, with a back stop and a side stop, so the strip cannot creep while a label is peeled off it.
6. One fresh label sheet lies flat on the fresh label spot. It holds well more than twelve labels, all of one kind, and never a mixed sheet.
7. The corner of the next fresh label is lifted off the sheet, so the gripper has an edge to close on. A label lying flat with no lifted corner is not usable: the gripper cannot lift the corner and peel the label at the same time.
8. One audit sheet lies flat on the audit sheet spot at the front left, with twelve numbered boxes, three across and four deep, all of them empty. The numbers match the position numbers on the strip.
9. Each audit box is wider and taller than a label, so a worn label lands inside its own box without touching the next one.

10. The waste spot at the front middle is empty or only part full. It takes bad fresh labels and nothing else.
11. One scan wand lies flat on the wand spot at the back right, window down, handle toward the front. It is light enough for the gripper to hold level and reads a label at a hand's width away.
12. A clear gap of at least a gripper's width sits around the cradle on every side, so the wand comes down over one position and lifts away without touching a cradle wall.

Workspace layout

Three rows: the fresh labels and the wand at the back, the strip in its cradle in the middle, the audit sheet and the waste spot at the front. Fresh labels travel forward to the strip, worn labels travel forward from the strip to the audit sheet.

Back row, left to right:

1. **Fresh label spot**, back left. The fresh label sheet lies flat here with the corner of the next label lifted. It empties as the audit sheet fills.
2. **Wand spot**, back right. The wand lies flat here, window down and handle toward the front.

Middle row:

1. **Cradle spot**, the middle of the bench. The cradle holds the strip flat with all twelve positions facing up. Everything that happens to a label happens here. The strip stays in the cradle for the whole session.

Front row, left to right:

1. **Audit sheet spot**, front left. The audit sheet lies flat here with its twelve numbered boxes. It is empty at the start of the session and holds twelve worn labels at the end, one in each box. It is the record of which positions were replaced.
2. **Waste spot**, front middle, the strip of bare bench between the cradle and the front edge. It takes bad fresh labels and nothing else.

No carry passes over a position other than the one being worked, no carry passes over the audit sheet on the way to somewhere else, and nothing travels to the far side of the bench.

Vocabulary

Shelf strip: the long strip taken off the shelf edge that carries the twelve labels. It is worked at the bench and never on the shelf.

Position: one of the twelve label patches on the strip, each with its own printed border and its own number. Positions run 1 to 12, left to right.

Position number: the number printed beside the position on the strip. It says which audit box the worn label goes in.

Worn label: a label that is faded, torn, scuffed, curled at a corner, or lifted off the strip, so the code on it is not fully readable. Every position starts with one.

Sound label: a label that is whole, flat, and fully readable. A sound label is left alone.

Bare position: the position with no label on it, its border and number in view, and nothing left behind but a flat patch.

Fresh label: one whole new label taken off the fresh label sheet. Labels are taken in sheet order, one per episode.

Bad fresh label: a fresh label that is torn, folded onto its own sticky side, or doubled with a second label. It goes to the waste spot.

Square: the label's long edges run parallel to the long edges of the position within 5 degrees, and the label sits inside the printed border on all four sides.

Press: the closed gripper held down on the label for 2 seconds, starting at the middle and working out to the edges.

Flat label: the whole label touching the patch, with no lifted corner, no crease, and no bubble under it.

Audit sheet: the sheet at the front left with twelve numbered boxes. One worn label goes in each box.

Audit box: one numbered box on the audit sheet. The box number matches the position number.

Replaced position: a position whose worn label is off, whose fresh label is on and scanned green, and whose worn label lies in its own audit box. Nothing less counts.

Scan wand: the handheld reader used to check the new label. It reads at a hand's width away and never touches the label.

Scan gap: visible daylight between the wand window and the label, held for the whole read.

Good read: one beep and a green light from the wand on the new label.

Cradle: the open holder that keeps the strip flat and still. It holds the strip, it never moves, and nothing else ever goes in it.

Hold: a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.

Tool home: the spot a tool returns to and lies flat on before the next step starts: the wand spot. A tool is never left on the strip, on the audit sheet or in the gripper between steps.

Release point: the gripper opens over a position, over an audit box, over the waste spot, or over a tool home. Never over another position, over the fresh label sheet or over bare bench.

Stable placement: the object lies still after release, flat on its surface, and does not rock, slide, curl up or lean on anything.

Loose label: a label that ends up on the bench outside a position, an audit box or the waste spot. It is left where it falls and cleared in the reset, never on camera.

Quarter turn: the most any object is turned in the gripper. Turning a label to line it up stays inside it. Nothing is ever turned over in the air.

Steps

Step 1: Look at the position and take the worn label off

Goal: the position is bare, its border and number in view, and the worn label lies flat in its own numbered box on the audit sheet.

The cradle grips the bench on its own, so nothing pins it while the worn label comes off.

1.1 Look at the position

Look at the next position in order, position 1 first, then each one to its right.

Read the position number and check that one worn label sits in the position.

Check: one worn label sits in the position and the position number is in view. If the label is sound, leave it where it is, end the episode, and fix the strip during reset. If the number cannot be read, end the episode and fix the strip during reset.

1.2 Peel the worn label off

Close on one corner of the worn label, at the corner nearest the front of the bench.

Peel it off in one move, up and toward the front, keeping the label held the whole way.

Lift it clear of the strip once the last of it comes away.

The label is peeled toward the front so the pull runs off the strip and not along it. A pull along the strip drags the strip against its stops.

Check: the position is bare, with nothing left behind on the patch, and one whole worn label hangs from the gripper. If part of the label stays stuck, close on that part and peel it once more. Make no more than two extra peels. If it still will not come away whole, lay what is held in its audit box, end the episode, and clear the patch during reset.

1.3 Lay the worn label in its box

Carry the worn label forward to the audit sheet, to the box with the same number as the position.

Lower it flat inside the box, open the gripper, and lift clear.

Check: the worn label lies flat and fully inside its own numbered box, touching no other box and no other label. If it lands across two boxes or curls up out of one, close on it, lift it clear, and lay it down again. Make no more than one correction.

Step 2: Peel one fresh label

Goal: exactly one fresh label hangs from the gripper, label face up and sticky side down, and the sheet still lies flat on its spot.

The strip stays in the cradle for this whole step. Nothing touches the strip while the fresh label is being taken.

Close on the lifted corner of the next label on the sheet.

Peel the label off in one move, up and toward the front of the bench, keeping the label face up the whole way.

Lift it clear of the sheet and hold it level.

Labels are taken in sheet order, the next one every time. A label is never skipped and never taken from somewhere else on the sheet.

Check: one whole fresh label hangs from the gripper, face up, and the sheet lies flat with no label half lifted. If the label tears, folds onto itself, or a second label comes away with it, lay it on the waste spot, open the gripper, and peel the next label once. Make no more than one extra peel.

Step 3: Place the fresh label square on the position

Goal: one fresh label lies flat and square inside the position's border, with no part of the patch showing through, no lifted corner and no crease.

3.1 Line the label up above the position

Carry the label to the strip and hold it a finger's width above the bare position, still clear of it.

Turn the label until its long edges run parallel to the long edges of the position, within 5 degrees.

Move it over until it sits inside the printed border on all four sides, with an even margin at the left and right.

The label is lined up in the air, before it touches anything. A label that touches the patch cannot be lined up again: the sticky side grabs at once.

Check: the label hangs square above the position, inside the border on all four sides. If it is turned or off to one side, hold it still and line it up again. Make no more than two corrections before it comes down.

3.2 Lower it and press it flat

Lower the label straight down onto the position until the whole sticky side is touching, then open the gripper and lift clear.

Close the gripper and press the label down for 2 seconds, starting at the middle and working out to the edges.

Lift the gripper clear of the label.

The press runs from the middle out so any air under the label is pushed off an edge. A press that starts at an edge traps a bubble in the middle.

Check: the label lies square inside the border, the whole label is touching, and no corner is lifted, creased or bubbled. If a corner is lifted, press that corner once more. Make no more than one extra press. If the label is off square or creased, leave it exactly where it is, end the episode, and strip the position during reset. A second label is never laid over the first one.

Step 4: Scan the new label

Goal: the wand has given one beep and a green light on the new label, and the wand is back flat on its spot.

The strip is not held while the scan runs. The wand comes to the label, and the label stays where it is.

4.1 Take the wand

Close on the wand handle, square to it, so the window hangs level and facing down.

Lift it straight up off the wand spot and carry it forward to the strip, level the whole way.

Bring the window over the middle of the new label, a hand's width above it, and stop.

The wand never passes over another position on its way in or out. It comes down over the position being worked and lifts straight off it.

4.2 Read the label

Hold the wand still at the scan gap until the wand beeps once and shows green.

If nothing reads, lift the wand, bring it back over the middle of the label at the same gap, and hold it still again.

Make no more than three tries in total.

Check: the wand has beeped once and shown green on the new label. If three tries give no read, park the wand, end the episode, and let review flag the position. The label is not peeled off and put on again to chase a read.

4.3 Park the wand

Carry the wand back to the wand spot, lower it until it is flat, window down and handle toward the front, then open the gripper and lift clear.

Check: the wand lies fully on its spot, flat, not draped over the cradle, the audit sheet or the bench edge.

Step 5: End the episode

Confirm the position: one fresh label lying flat and square inside the border, fully pressed.

Confirm the read: the wand beeped once and showed green on that label.

Confirm the audit sheet: the worn label lying flat in the box with the same number as the position, and every box up to this position holding one worn label.

Confirm the strip: still flat in the cradle, on both stops, and every position to the right of this one still carrying its worn label.

Confirm the tools: the wand flat on its spot, the fresh label sheet flat on its spot with the next corner lifted.

Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.

Go home with the gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the corner of the next fresh label on the sheet, so the next episode has an edge to close on.
2. Confirm the audit sheet holds one worn label in every box up to the position just done, each flat and inside its own box. Flatten any that has curled.
3. Pick any loose label off the bench and put it in the bin. Wipe and dry the patch it was stuck to.
4. Empty the waste spot into the bin if it is more than part full.
5. Confirm the strip lies flat in the cradle, on both stops, and that the cradle still grips the bench.
6. Confirm the position just done still holds its fresh label flat and square, and that the next position still holds its worn label.
7. Confirm the wand still reads a spare fresh label with one beep and a green light. Charge or replace a wand that does not.
8. Run both Setup checklists again.

At the end of a session, once all twelve positions are done: take the audit sheet with its twelve worn labels and file it as the record of the replaced positions, put a fresh strip in the cradle with a fresh audit sheet, and put the finished strip back on the shelf edge. None of this is recorded.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

Visible cue: a fresh label is peeled before the position is bare, the worn label is still held when the fresh label is taken, the label is pressed before it is lined up, the wand comes over the label before it is pressed, the worn label goes into its box after the scan instead of before the fresh label is peeled, positions are worked out of order, or a step is skipped.

SOP rule broken: Steps 1 to 4, look, peel off, box it, peel fresh, place, press, scan, in that order, positions worked left to right.

Coaching note: old one off and boxed, new one on, then read it. The position is bare before a fresh label leaves the sheet.

Violation: Fresh label put on a label

Visible cue: the fresh label is lowered onto a position that still carries all or part of the worn label, or a second fresh label is laid over a first one.

SOP rule broken: Steps 1.2 and 3, the position is bare before a fresh label comes down on it, and one label per position.

Coaching note: one label on a bare patch. A label over a label hides the fault instead of fixing it.

Violation: Sound label replaced

Visible cue: the label peeled off the position was whole, flat and fully readable.

SOP rule broken: Step 1.1, only a worn label is replaced, and a sound label is left alone.

Coaching note: look before peeling. A sound label is a reset error to report, not work to do.

Violation: Worn label boxed wrong, or not boxed

Visible cue: the worn label is put in a box whose number does not match the position, is laid across two boxes, is dropped on the bench or in the waste spot, or is still in the gripper when the fresh label is taken.

SOP rule broken: Step 1.3, the worn label goes flat inside the box with the same number as the position.

Coaching note: the audit sheet is the record of the replaced positions. A label in the wrong box says the wrong position was done.

Violation: Peel done wrong

Visible cue: the worn label is pulled along the strip instead of toward the front, is scrubbed or scraped at with the gripper, is torn into pieces left on the patch, or the peel drags the strip off its stops.

SOP rule broken: Step 1.2, close on the front corner and peel up and toward the front in one move.

Coaching note: one corner, one move, toward the front. The pull comes off the strip, never along it.

Violation: Label not square

Visible cue: the label lands turned more than 5 degrees from the position's long edges, crosses the printed border on any side, or hangs off the patch.

SOP rule broken: Steps 3.1 and 3.2, the label is lined up square and inside the border before it comes down.

Coaching note: line it up in the air. Once it touches, it is where it is going to stay.

Violation: Label lined up on the patch

Visible cue: the label touches the position and is then turned, slid or lifted and put down again to line it up.

SOP rule broken: Step 3.1, the label is lined up above the position, clear of it.

Coaching note: the sticky side grabs on contact. All the aiming happens in the air.

Violation: Press missing or wrong

Visible cue: the gripper opens and leaves with no press, the press is visibly shorter than 2 seconds, the press starts at an edge and works in, or a corner is left lifted, creased or bubbled.

SOP rule broken: Step 3.2, press for 2 seconds from the middle out until the whole label is touching.

Coaching note: middle first, then out to the edges. That is what pushes the air off the edge.

Violation: Bad fresh label used

Visible cue: a fresh label that is torn, folded onto its own sticky side, or doubled is carried to the position and put down on it.

SOP rule broken: Step 2, a bad fresh label goes to the waste spot and the next label is peeled.

Coaching note: a bad label is waste. Lay it down and take the next one.

Violation: Wrong fresh label taken, or more than one

Visible cue: a label is skipped on the sheet, a label is taken from somewhere else on the sheet, or two labels come away and both are carried to the position.

SOP rule broken: Step 2, take the labels in sheet order, one label per episode.

Coaching note: the next label on the sheet, every time.

Violation: Scan skipped, or a bad read passed

Visible cue: the episode ends with no scan of the new label, or it ends after the wand gave no beep and no green light.

SOP rule broken: Step 4, every new label is read, with one beep and a green light.

Coaching note: the read is what makes the position replaced. No read, no pass.

Violation: Scan done wrong

Visible cue: the wand window touches the new label or rests on the strip, the wand is held over the border or over another position instead of the middle of the label, the wand is tilted or turned over, or it is swept back and forth over the label instead of held still.

SOP rule broken: Steps 4.1 and 4.2, bring the window over the middle of the label at the scan gap and hold it still.

Coaching note: set the gap over the middle, then hold. The wand reads from a distance, so it never needs to land.

Violation: Scan tries over the limit

Visible cue: a fourth try is made on the same label, or the label is peeled off and put on again to chase a read.

SOP rule broken: Step 4.2, no more than three tries, and a failed read is flagged rather than chased.

Coaching note: three tries, then park the wand and let review flag it.

Violation: Strip or cradle moved

Visible cue: the strip lifts, slides or comes out of the cradle; the cradle is pushed, nudged or lifted; the strip is held up in the air for a peel, a press or a scan; or the strip comes off one stop or both stops.

SOP rule broken: Steps 1 to 4, the strip stays flat in the cradle on both stops for the whole session, and the cradle is never moved.

Coaching note: the strip is worked in place. Nothing on this bench needs picking up except the labels and the wand.

Violation: Loose label cleaned up

Visible cue: after a label lands on the bench outside a position, an audit box or the waste spot, the gripper is used to sweep, push, wipe or pick it up.

SOP rule broken: Steps 1 to 3, a loose label is left where it falls and is cleared in the reset.

Coaching note: cleanup motions are not part of the task. The loose label is data, the cleanup is noise.

Violation: A carry passes over another position

Visible cue: a fresh label, a worn label or the wand passes over a position other than the one being worked, or a carry passes over the audit sheet on the way to somewhere else.

SOP rule broken: Steps 1.3, 2, 3.1 and 4.1, every carry goes around the other positions, never over them.

Coaching note: route around, not across. A dropped label over the wrong position ruins two records at once.

Violation: Bad pick, carry, or release

Visible cue: an object moves from one grip to another without being set down first, a grip is shifted or seated again while the object is off the bench, the wand or a label slides along the bench with no daylight under it, the wand hangs tilted because the handle grasp was taken off square, or an object is turned past a quarter turn or turned over in the air.

SOP rule broken: Steps 1 to 4, one object at a time, lifted straight up clear of the surface before any sideways motion, gripped square, turned no more than a quarter turn, and never passed between grips.

Coaching note: set it down to change the grip, and lift before the carry. A wand that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

Visible cue: a label, the strip, the audit sheet or the wand is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.

SOP rule broken: Steps 1 to 4, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.

Coaching note: nothing on this bench is pushed into line. If the grip or the approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

Visible cue: the gripper opens while the wand is still above its spot, the wand is left on the strip, on the audit sheet or on bare bench, the wand is laid down window up or handle away, or the wand is still held when the episode ends.

SOP rule broken: Step 4.3, the wand returns to its spot and lies flat there, window down and handle toward the front, before the episode ends.

Coaching note: the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Dropped, collided, or knocked over

Visible cue: under command, the gripper or a carried object strikes the cradle, the strip, the audit sheet, the label sheet or the wand and knocks, tips, slides or displaces it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.

SOP rule broken: Steps 1 to 4, keep each object on a clear travel path and release only after a settled placement.

Coaching note: watch the whole path, not just the target.

Violation: Manipulation outside the camera frame

Visible cue: a peel, a placement, a press or a scan happens partly or wholly outside the environment camera frame.

SOP rule broken: Steps 1 to 4, every manipulation happens inside the camera frame.

Coaching note: the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

Visible cue: the episode ends with the position bare or still carrying its worn label, the fresh label unpressed or unread, the worn label out of its box, the strip out of the cradle, the wand out of its spot, short of home, or the gripper not fully open.

SOP rule broken: Step 5, confirm the position, the read, the audit sheet, the strip and the tools, then go home with the gripper open and stop recording.

Coaching note: confirm first. Homing is the last thing that happens before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Hardware fault: gripper failure, drift, controller caused collision, or motor error.

A spot sits outside comfortable reach, so the move stalls at a joint limit or has to fold in on itself to work the spot. Move the spot and run the Setup checklists again.

Defective cradle: one that slides or lifts under a correct peel, or that will not hold the strip flat and on both stops.

Defective strip: one whose patch is torn or greasy so a fresh label will not lie down on it, whose position numbers cannot be read, or that will not sit flat in the cradle.

Defective label sheet: one whose labels will not peel away whole under a correct peel, one with no lifted corner that can be made, or one whose labels have dried out and will not stick.

Defective wand: one that gives no beep or no light on a known good spare label, whose battery is flat, or that only reads with the window touching the label.

Worn label bonded so hard that it will not come away whole under a correct peel and two extra peels. Material issue, not a violation. Strip the patch during reset.

Defective audit sheet: one whose boxes are smaller than a label, or whose numbers do not match the strip.

Fewer than twelve fresh labels on the sheet, or fewer than twelve empty boxes on the audit sheet. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Look at the position and read its number
2. Peel the worn label off the position
3. Lay the worn label in its audit box
4. Peel one fresh label from the sheet
5. Line the fresh label up above the position
6. Lower the fresh label onto the position
7. Press the fresh label flat
8. Take the wand from its spot
9. Scan the new label
10. Return the wand to its spot
11. Return home and end the episode